

Eta Cassiopeiae
1 Sun

Caph
30 Suns

Ruchbah
70 Suns

CASSIOPEIA

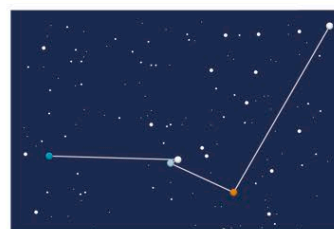
THE VAIN QUEEN

CASSIOPEIA LIES WITHIN THE BAND OF THE MILKY WAY. ITS FIVE MAIN STARS FORM A ZIGZAG SHAPE RESEMBLING THE LETTER "W" THAT MAKES THIS CONSTELLATION EASY TO RECOGNIZE IN THE NORTHERN SKY.

Cassiopeia was a vain queen of Greek mythology, the wife of King Cepheus. As punishment for Cassiopeia's vanity, the sea god Poseidon sent a monster to ravage her country's coastline. To rid themselves of the monster, Cassiopeia and Cepheus chained their daughter Andromeda to a rock as a sacrifice. Fortunately, she was rescued from the monster's jaws by the hero Perseus. All the characters in this myth are represented by constellations close together in the night sky.

Cassiopeia contains the remains of two supernova explosions. One, called Tycho's Star, became visible from Earth in 1572. The other occurred about a century later, but went unseen at the time.

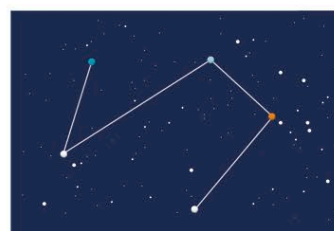
The major features of the constellation for users of small telescopes are the beautiful double star Eta Cassiopeiae and several open clusters of stars, notably M52, M103, and NGC 457.



50,000 BCE



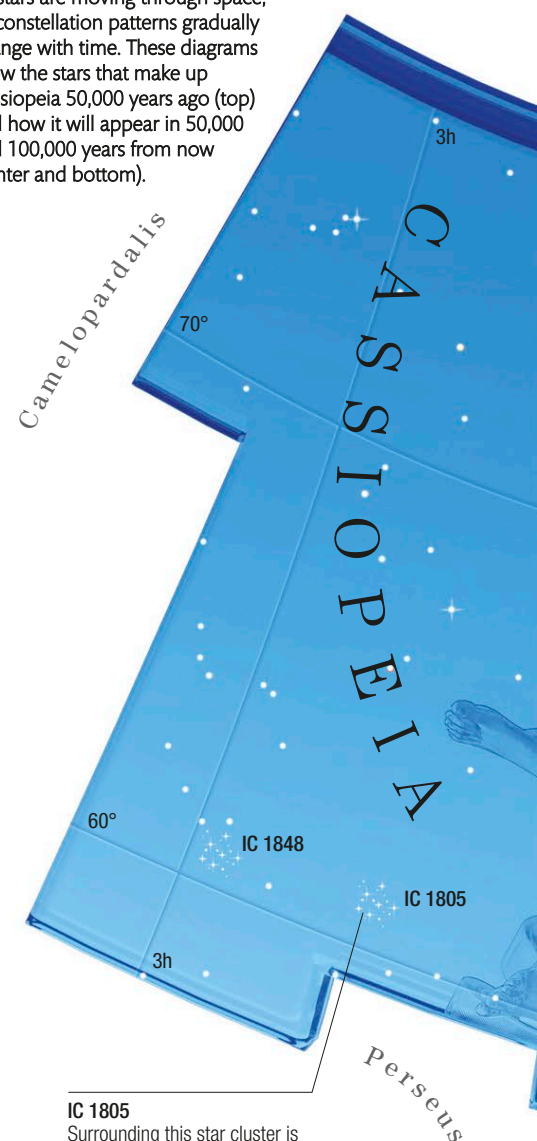
50,000 CE



100,000 CE

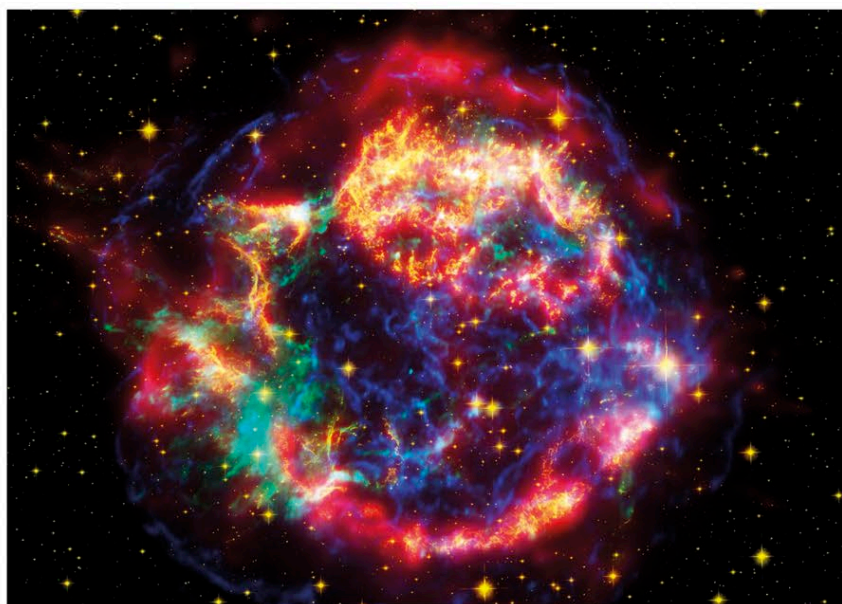
◀ Changing shape

All stars are moving through space, so constellation patterns gradually change with time. These diagrams show the stars that make up Cassiopeia 50,000 years ago (top) and how it will appear in 50,000 and 100,000 years from now (center and bottom).



◀ Cassiopeia A supernova remnant

The strongest radio source in the sky, Cassiopeia A has been identified as the remains of a supernova explosion some 11,000 light-years away. Light from the supernova should have reached Earth in the 1600s. However, there is no record of it having been observed, so it was probably dimmed by the surrounding dust. This image of the exploded star is a composite of observations made at infrared (red), optical (yellow), and X-ray (green and blue) wavelengths.



IC 1805

Surrounding this star cluster is a cloud of glowing gas called the Heart Nebula, so-named because it resembles the human heart in shape.

▷ Star distances

One might be forgiven for thinking that the five main stars in Cassiopeia's distinctive "W" formation are relatively close together, but in fact they lie at greatly differing distances from Earth. The farthest away, Gamma Cassiopeiae (the central star in the "W"), is more than ten times more distant than the nearest of the five to us, Caph (Beta Cassiopeiae).

